Operator-norm convergence of unscaled subdiagonal Padé approximants for bounded analytic semigroups

Candidate full solution for arXiv:1210.8408, Remark 4.7

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Status. Claimed full solution, likely valid, pending expert review. The result proves the stronger operator-norm conclusion asked about in the source paper, for every bounded analytic semigroup on a complex Banach space.

1 Source question

Egert and Rozendaal study the first subdiagonal Padé approximants $r_n = [n/(n+1)]$ to the exponential [1]. Their Theorem 4.6 proves, for a bounded analytic semigroup $T(t) = e^{-tA}$, convergence on every fractional domain $D(A^\alpha)$, $\alpha > 0$. They then ask in Remark 4.7 (arXiv PDF p. 13) whether the method *without scaling and squaring* converges strongly on the whole space and even in operator norm. The answer is affirmative. In fact, the convergence below is uniform in $t \geq 0$.

The rightmost term can be estimated by means of Lemma 3.5. This leads to

$$\|(r_n(-A) - T(1))\, x\| \leq \frac{4M_\nu}{\alpha\pi}(n+1)^{-\alpha}\|A^\alpha x\|,$$

which is the required estimate for $t = 1$. □

Remark 4.7. In the situation of Theorem 4.6 the scaling and squaring methods associated to a fixed subdiagonal Padé approximant converge strongly on X and even in $\mathcal{L}(X)$, see [15, Thm. 4.4]. Whether this is true for the method without scaling and squaring as well is left as an open problem. In Section 4.4 we will prove strong convergence on X under an additional assumption on A.

Figure 1: The open problem in Egert–Rozendaal, Remark 4.7, p. 13.

2 Notation and theorem

Let A be sectorial of angle $\varphi < \pi/2$ on a complex Banach space X, and let $-A$ generate the bounded analytic semigroup $(e^{-tA})_{t\geq 0}$. For $\nu \in (\varphi, \pi/2)$ put

$$M_\nu := \sup_{z\in\partial\Sigma_\nu} \|z(z-A)^{-1}\|, \qquad \delta_\nu := 1 - \frac{2\nu}{\pi} > 0,$$

where $\Sigma_\nu = \{z \neq 0 : |\arg z| < \nu\}$. The Padé approximant $r_n = P_n/Q_n$ is normalized by $P_n(0) = Q_n(0) = 1$, with $\deg P_n = n$ and $\deg Q_n = n+1$, and satisfies $r_n(z) - e^z = O(z^{2n+2})$ at zero.

Theorem 1 (Full operator-norm endpoint)**.** *For every $\nu \in (\varphi, \pi/2)$ there is a constant C_ν such that, for all $n \geq 1$,*

$$\sup_{t\geq 0}\left\|r_n(-tA) - e^{-tA}\right\| \leq C_\nu M_\nu (n+1)^{-\delta_\nu}\bigl(1 + \log(n+1)\bigr).$$

Consequently $r_n(-tA) \to e^{-tA}$ in $\mathcal{L}(X)$, uniformly for $t \geq 0$. In particular, both conclusions asked for in Remark 4.7 of [1] hold without any additional bounded H^∞-calculus assumption.

3 Proof intuition

The obstruction at the endpoint $\alpha = 0$ is logarithmic: the crude $\mathcal{A}$-stability estimate $|r_n(-z)-e^{-z}| \leq 2$ is not integrable against $|dz|/|z|$ on a sectorial contour. A classical theorem of Saff–Varga–Ni supplies the missing boundary information: the unscaled $[n/(n+1)]$ Padé error is at most $1/(2(n+1))$ on the positive real axis. The two-constants theorem transfers this decay to every strict sector, with exponent $\delta_\nu = 1 - 2\nu/\pi$.

Uniform sector decay alone still leaves an infinite logarithmic interval. The second ingredient is a polynomial tail bound $|r_n(-z)| \leq 4(n+1)^2/|z|$ in the right half-plane. It follows from an exact relation between P_n and Q_n, followed by the Eneström–Kakeya root bound. Splitting the contour into a small piece, a middle piece, and a tail then gives $O((n+1)^{-\delta_\nu}\log(n+1))$. The holomorphic functional calculus turns this scalar estimate into the operator-norm estimate.

4 Scalar estimates

Set

$$N := n+1, \qquad F_n(z) := r_n(-z) - e^{-z} \quad (\operatorname{Re} z \geq 0).$$

Lemma 2 (Positive-ray and strict-sector estimates)**.** *For $n \geq 1$,*

$$\sup_{x\geq 0}|F_n(x)| \leq \frac{1}{2N}.$$

Moreover, for $0 \leq \theta < \pi/2$, $r \geq 0$, and $\delta_\theta := 1 - 2\theta/\pi$,

$$|F_n(re^{\pm i\theta})| \leq 2N^{-\delta_\theta}.$$

Proof. Write $R_{\mu,m}$ for the type (μ, m) Padé approximant to e^{-z} in the notation of Saff–Varga–Ni. Their Theorem 2.1 gives

$$\sup_{x\geq 0}|e^{-x} - R_{\mu,m}(x)| \leq \frac{1}{2^{m-\mu}\binom{m}{\mu}} \qquad (0 \leq \mu \leq m)$$

[2, Theorem 2.1]. Taking $(\mu, m) = (N-1, N)$ gives the first claim.

The function F_n is bounded and holomorphic in the right half-plane and is continuous on its closure. On the positive boundary ray it is bounded by $(2N)^{-1}$, while $\mathcal{A}$-stability gives $|F_n(iy)| \leq 2$ on the imaginary boundary ray. The two-constants theorem in the first quadrant yields

$$|F_n(re^{i\theta})| \leq (2N)^{-\delta_\theta}2^{1-\delta_\theta} \leq 2N^{-\delta_\theta}.$$

The lower quadrant follows by conjugation. $\square$

Lemma 3 (Polynomial tail)**.** *For $n \ge 1$ and $\operatorname{Re} z \ge 0$, $z \ne 0$,*

$$|r_n(-z)| \le \frac{4N^2}{|z|}.$$

Proof. Using the explicit Padé coefficients from [1], write

$$Q_n(z) = \sum_{j=0}^{N} q_j (-z)^j, \qquad q_j = \frac{(2N-1-j)!N!}{(2N-1)!j!(N-j)!} > 0.$$

If p_j is the coefficient of z^j in P_n, direct cancellation gives

$$\frac{p_j}{q_j} = \frac{N-j}{N} \qquad (0 \le j \le N-1).$$

Consequently

$$P_n(-z) = Q_n(z) - \frac{z}{N} Q_n'(z). \tag{1}$$

Apply the Eneström–Kakeya theorem to $Q_n(-z) = \sum_{j=0}^{N} q_j z^j$. Every zero λ of Q_n satisfies

$$|\lambda| \le \max_{0 \le j < N} \frac{q_j}{q_{j+1}} = \max_{0 \le j < N} \frac{(2N-1-j)(j+1)}{N-j} \le 2N^2.$$

Let $\lambda_1, \dots, \lambda_N$ be the roots of Q_n. Since Q_n has real coefficients, (1) and logarithmic differentiation show, for $t \in \mathbb{R}$, that

$$\begin{aligned} |r_n(-it)| &= \left| 1 - \frac{it}{N} \frac{Q_n'(it)}{Q_n(it)} \right| \\ &= \left| \frac{1}{N} \sum_{k=1}^{N} \frac{\lambda_k}{\lambda_k - it} \right|. \end{aligned}$$

If $|t| \le 4N^2$, $\mathcal{A}$-stability gives $|t r_n(-it)| \le 4N^2$. If $|t| \ge 4N^2$, then $|\lambda_k - it| \ge |t|/2$, and the displayed formula again gives $|t r_n(-it)| \le 4N^2$. Thus $h_n(z) := z r_n(-z)$ is bounded by $4N^2$ on the imaginary axis. It is holomorphic on the closed right half-plane and has a finite limit at infinity. The maximum principle on the compactified half-plane gives $|h_n(z)| \le 4N^2$ throughout $\operatorname{Re} z \ge 0$. □

Lemma 4 (Contour integral)**.** *Fix $0 < \nu < \pi/2$ and put $\delta = 1 - 2\nu/\pi$. Then*

$$I_{n,\nu} := \int_0^\infty |F_n(re^{i\nu})| \frac{dr}{r} \le C_\nu N^{-\delta}(1 + \log N).$$

The same estimate holds on the ray of angle $-\nu$.

Proof. Egert–Rozendaal's Perron estimate gives, for $\operatorname{Re} z \ge 0$,

$$|F_n(z)| \le a_n |z|^{2N}, \qquad a_n := \frac{1}{2}\left(\frac{n!}{(2n+1)!}\right)^2 \le \frac{1}{2} \quad [1, Lemma 3.2]. \tag{2}$$

Let $R = N^{2+\delta}$. On $(0,1)$ use (2); on $(1,R)$ use Lemma 2; and on (R,∞) use Lemma 3 and $|e^{-re^{i\nu}}| = e^{-r\cos\nu}$. This gives

$$\begin{aligned} I_{n,\nu} &\le \frac{a_n}{2N} + 2N^{-\delta} \log R + \int_R^\infty \left(\frac{4N^2}{r} + e^{-r\cos\nu} \right) \frac{dr}{r} \\ &\le \frac{1}{4N} + 2(2+\delta) N^{-\delta} \log N + 4N^{-\delta} + \frac{e^{-R\cos\nu}}{R\cos\nu}. \end{aligned}$$

Since $0 < \delta < 1$, this is bounded by $C_\nu N^{-\delta}(1 + \log N)$. Conjugation handles the lower ray. □

5 Proof of Theorem 1

The case $t = 0$ is exact. Fix $t > 0$. The function

$$G_{n,t}(z) := r_n(-tz) - e^{-tz} = F_n(tz)$$

belongs to the standard H_0^∞ class on every sector strictly contained in the right half-plane: it vanishes to order $2N$ at zero and is $O(|z|^{-1})$ at infinity there. The sectorial functional calculus gives

$$r_n(-tA) - e^{-tA} = \frac{1}{2\pi i} \int_{\partial \Sigma_\nu} G_{n,t}(z)(z - A)^{-1}\, dz.$$

Taking norms and using the definition of M_ν yields

$$\|r_n(-tA) - e^{-tA}\| \le \frac{M_\nu}{2\pi} \sum_{\epsilon \in \{-1,1\}} \int_0^\infty |F_n(tre^{i\epsilon\nu})| \frac{dr}{r}.$$

The substitution $s = tr$ removes t. Lemma 4 proves

$$\|r_n(-tA) - e^{-tA}\| \le C_\nu M_\nu N^{-\delta_\nu}(1 + \log N),$$

uniformly for $t > 0$, which is the assertion.

6 Verification and limitations

The included script `code/verify_pade_endpoint.py` checks the exact coefficient identity for $1 \le n \le 20$, verifies the coefficient-ratio root bound for $1 \le n \le 200$, and numerically samples the positive-ray, imaginary-tail, and strict-sector integral estimates. These computations are sanity checks only; the proof above is analytic.

The conclusion is exactly scoped to the open problem's bounded-analytic setting. It does not settle strong convergence on all of X for an arbitrary bounded, nonanalytic C_0-semigroup; that broader problem is still described as open in the 2024 work of Gomilko–Tomilov [4].

The proof deliberately uses a coarse $O(N^2/|z|)$ rational tail. Numerical experiments suggest the sharper bound $O(N/|z|)$ and a faster contour rate, but neither sharpening is needed for convergence and neither is claimed here.

7 Novelty check and review recommendation

A bounded search covered the exact Remark 4.7 wording, the core subdiagonal-Padé/operator-norm/bounded-analytic-semigroup phrases, arXiv:1210.8408 citations, and the later papers of Neubrander–Özer–Windsperger [3], Gomilko–Tomilov [4], and Batty–Gomilko–Tomilov [5]. No matching theorem for the unscaled variable sequence $[n/(n+1)]$ on arbitrary Banach-space bounded analytic semigroups was found. The 2024/2025 works address either the general bounded-semigroup problem on Hilbert spaces with positive regularity, or fixed-rational-function scaling formulas $r(tA/n)^n$.

Human-review recommendation. Verify first the Saff–Varga–Ni indexing $(\mu, m) = (n, n+1)$, then (1) and the maximum-principle passage in Lemma 3. If these pass, the remaining contour-calculus argument is standard.

References

[1] M. Egert and J. Rozendaal, *Convergence of subdiagonal Padé approximations of C_0-semigroups*, J. Evol. Equ. **13** (2013), 875–895; arXiv:1210.8408.

[2] E. B. Saff, R. S. Varga, and W.-C. Ni, *Geometric convergence of rational approximations to e^{-z} in infinite sectors*, Numer. Math. **26** (1976), 211–225.

[3] F. Neubrander, K. Özer, and L. Windsperger, *On subdiagonal rational Padé approximations and the Brenner–Thomée approximation theorem for operator semigroups*, Discrete Contin. Dyn. Syst. S **13** (2020), 3565–3579.

[4] A. Gomilko and Y. Tomilov, *Rational approximation of operator semigroups via the $\mathcal{B}$-calculus*, J. Funct. Anal. **287** (2024), 110426; arXiv:2403.14411.

[5] C. Batty, A. Gomilko, and Y. Tomilov, *Rational approximation of holomorphic semigroups revisited*, J. Lond. Math. Soc. **111** (2025), e70066; arXiv:2403.15894.